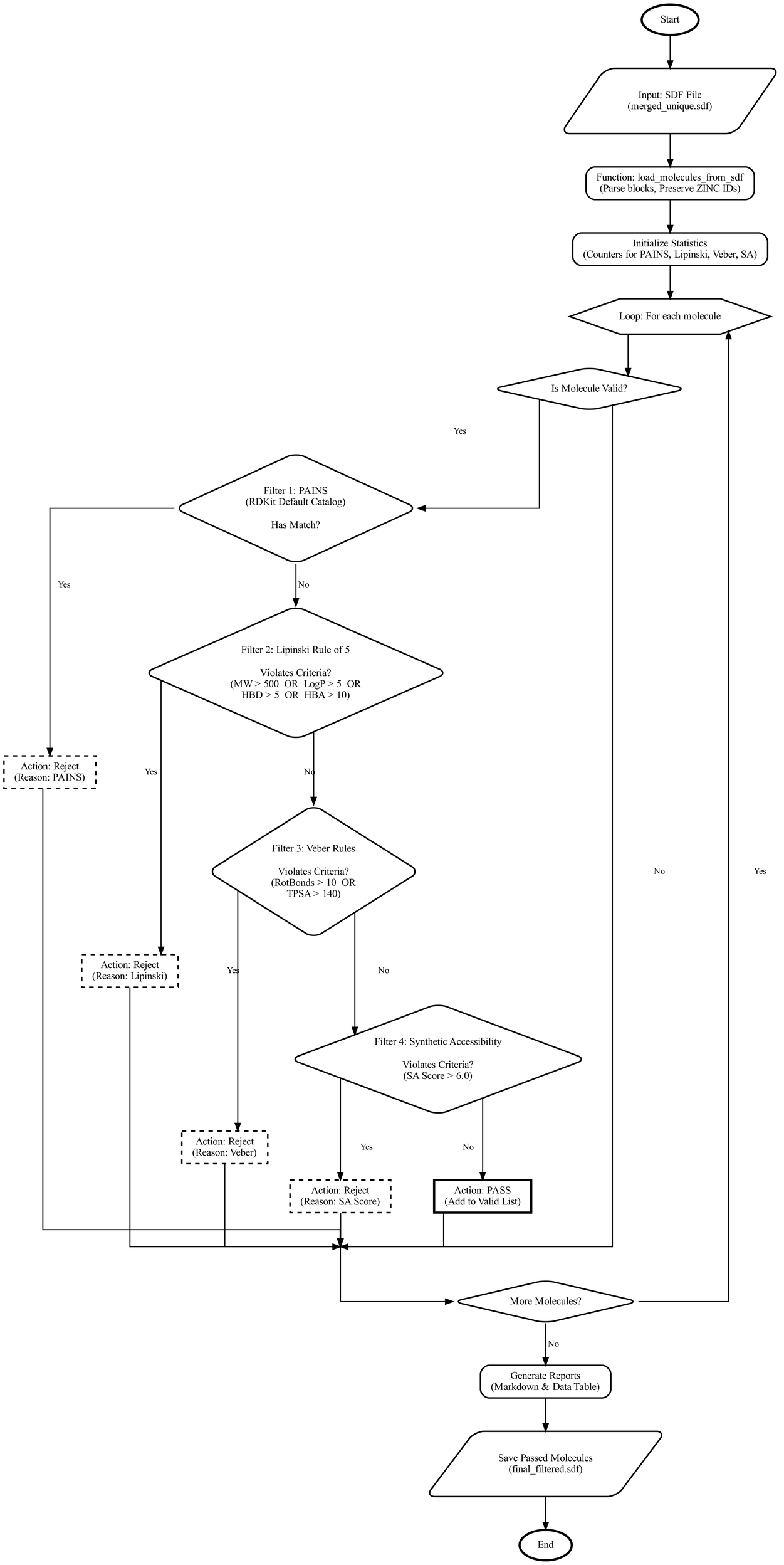
Start
Input: SDF File
(merged_unique.sdf)
Function: load_molecules_from_sdf
(Parse blocks, Preserve ZINC IDs)
Initialize Statistics
(Counters for PAINS, Lipinski, Veber, SA)
Loop: For each molecule
Is Molecule Valid?
Yes
No
Yes
Filter 1: PAINS
(RDKit Default Catalog)
Has Match?
Yes
No
Action: Reject
(Reason: PAINS)
Filter 2: Lipinski Rule of 5
Violates Criteria?
(MW > 500 OR LogP > 5 OR
HBD > 5 OR HBA > 10)
Yes
No
Action: Reject
(Reason: Lipinski)
Filter 3: Veber Rules
Violates Criteria?
(RotBonds > 10 OR
TPSA > 140)
Yes
No
Action: Reject
(Reason: Veber)
Filter 4: Synthetic Accessibility
Violates Criteria?
(SA Score > 6.0)
Yes
No
Action: Reject
(Reason: SA Score)
Action: PASS
(Add to Valid List)
More Molecules?
No
Yes
Generate Reports
(Markdown & Data Table)
Save Passed Molecules
(final_filtered.sdf)
End